

Sales Analytics & Customer Performance Reporting Using Excel

Executive Summary

Organizations generate vast amounts of sales data every day, making it challenging to extract meaningful insights for decision-making. This project was developed to transform raw sales transactions into actionable business intelligence through data modeling, ETL processes, and analytical reporting using Microsoft Excel.

The project focuses on analyzing sales performance across customers, products, divisions, and global markets for AtliQ Hardware. By leveraging Power Query, Power Pivot, and dimensional modeling techniques, a scalable reporting solution was developed that enables stakeholders to monitor business performance, identify growth opportunities, and make data-driven decisions.

The final solution processes nearly 800,000 sales records and delivers multiple analytical reports, including Customer Performance Analysis, Market Performance Analysis, Division Performance Analysis, and Product Performance Analysis.

1. Introduction

Sales analytics plays a critical role in helping organizations understand customer behavior, market trends, product performance, and revenue growth. Traditional reporting methods often struggle to handle large datasets efficiently and lack the flexibility required for multidimensional analysis.

This project demonstrates the implementation of a Business Intelligence solution using Microsoft Excel to address these challenges. Through a combination of ETL processes, data modeling, and advanced reporting techniques, the project provides a centralized platform for analyzing organizational sales performance.

2. Business Problem Statement

AtliQ Hardware operates across multiple countries and serves a large customer base through various distribution channels. As sales volumes increased, management required a reporting framework capable of:

- Monitoring revenue performance across years.
- Evaluating customer growth and contribution.
- Identifying top-performing markets.
- Measuring division-wise business performance.
- Tracking product-level growth trends.
- Supporting strategic planning with reliable business insights.

The absence of a centralized analytical model made it difficult to perform detailed business analysis and generate consistent reports.

3. Project Objectives

The primary objectives of this project were:

Data Preparation

- Clean and transform raw sales data.
- Handle missing and inconsistent records.
- Standardize data formats.

Data Modeling

- Build a dimensional data model.
- Establish relationships between transactional and master data.
- Implement a Star Schema architecture.

Business Reporting

- Develop customer performance reports.
- Analyze market-wise sales trends.
- Evaluate division-level growth.
- Identify top-performing products.

Business Insights

- Discover revenue drivers.
- Identify high-growth markets.
- Support strategic decision-making.

4. Dataset Description

The project utilizes a sales dataset containing transactional and master data.

Fact Table

fact_sales_monthly

Contains monthly sales transactions used for revenue analysis.

Total Records: **799,962**

Dimension Tables

dim_customer

Contains customer information.

Total Records: **189**

dim_product

Contains product-related attributes.

Total Records: **298**

dim_market

Contains market and geographic information.

Total Records: **23**

dim_date

Custom Date Dimension created using Power Query.

Total Records: **1,066**

5. Technology Stack

The project was developed entirely within Microsoft Excel using the following technologies:

Tool	Purpose
Excel	Reporting & Analysis
Power Query	ETL Operations
Power Pivot	Data Modeling
DAX	Calculations & Measures
Pivot Tables	Data Aggregation
Conditional Formatting	Visual Insights

6. ETL Process

Extract

Raw sales and dimension tables were imported into Power Query from source datasets.

Transform

Several transformation operations were performed:

Data Cleaning

- Removed null values
- Handled missing records
- Standardized column formats

- Corrected inconsistencies
- Verified data quality

Data Transformation

- Renamed columns
- Applied appropriate data types
- Generated derived attributes
- Created a Date Dimension table

Date Table Creation

A separate Date Dimension was developed using Power Query to support time-based analysis and reporting.

The Date table includes:

- Date
- Month
- Quarter
- Year
- Fiscal Period

Load

After transformation, all tables were loaded into the Excel Data Model for analytical reporting.

7. Data Modeling

A Star Schema architecture was implemented to improve performance and maintain analytical flexibility.

Fact Table

fact_sales_monthly

Dimension Tables

- dim_customer
- dim_product
- dim_market
- dim_date

The dimensional model enables efficient filtering and slicing of sales data across multiple business perspectives.

Benefits

- Improved report performance
- Simplified analytical queries
- Enhanced scalability
- Better maintainability

8. Report Development

Five business-focused reports were developed.

8.1 Customer Performance Report

This report evaluates customer-level sales performance across multiple years.

Metrics

- Net Sales 2019
- Net Sales 2020
- Net Sales 2021
- Growth Percentage

Key Findings

Top Customers in 2021:

Customer	Sales
Amazon	\$82.1M
AtliQ Exclusive	\$61.1M
AtliQ e Store	\$53.0M

The report highlights significant customer growth and identifies major revenue contributors.

8.2 Division Performance Report

This report measures revenue contribution and growth across divisions.

Results

Division Net Sales 2021

P & A \$338.4M

Division Net Sales 2021

PC \$165.8M

N & S \$94.7M

Insights

- PC Division recorded the highest growth rate.
- P & A Division generated the largest revenue contribution.

8.3 Market Performance Report

This report evaluates business performance across global markets.

Top Markets

Market	Sales 2021
India	\$161.3M
USA	\$87.8M
South Korea	\$49.0M
Canada	\$35.1M
United Kingdom	\$34.2M

Insights

India emerged as the strongest market, contributing the highest revenue to the organization.

8.4 India Market Analysis

A dedicated report was created to analyze customer performance within India.

Leading Customers

- Amazon
- AtliQ Exclusive
- Flipkart
- Reliance Digital
- Vijay Sales

Insights

The Indian market demonstrated exceptional growth and represented the company's largest revenue-generating region.

8.5 Top Product Analysis

This report identifies products with the strongest growth and business impact.

High-Growth Products

- AQ Mx NB
- AQ Smash 2
- AQ LION x3
- AQ LION x2
- AQ LION x1

Insights

Several products recorded exponential year-over-year growth, indicating strong market demand and product acceptance.

9. Key Business Insights

Revenue Expansion

The organization achieved substantial growth between 2020 and 2021, reflecting successful market expansion initiatives.

Customer Concentration

A small group of customers contributed a significant portion of overall revenue, highlighting the importance of customer relationship management.

Market Leadership

India established itself as the most valuable market for the organization.

Product Growth

Several product categories experienced exceptional sales growth and represent future revenue opportunities.

Division Performance

The PC division demonstrated the strongest growth trajectory among all business units.

10. Business Recommendations

Market Strategy

Increase investments in high-performing markets such as India and the United States.

Customer Strategy

Strengthen partnerships with key customers including Amazon and AtliQ Exclusive.

Product Strategy

Expand production and marketing efforts for high-growth products.

Operational Strategy

Utilize analytical reporting for ongoing performance monitoring and strategic planning.

11. Project Outcomes

- Processed and analyzed nearly 800,000 sales records.
- Developed a complete ETL workflow using Power Query.
- Built a Star Schema data model using Power Pivot.
- Created multiple business-focused analytical reports.
- Generated actionable insights across customers, products, markets, and divisions.
- Demonstrated practical Business Intelligence implementation using Microsoft Excel.

12. Skills Demonstrated

Data Engineering

- ETL Development
- Data Cleaning
- Data Transformation

Data Modeling

- Star Schema Design
- Relationship Modeling
- Dimensional Modeling

Analytics

- Sales Analytics
- Customer Analytics
- Product Analytics
- Market Analysis

Business Intelligence

- KPI Reporting
- Dashboard Development

- Performance Monitoring

Tools

- Microsoft Excel
- Power Query
- Power Pivot
- DAX
- Pivot Tables

Conclusion

This project demonstrates the complete implementation of a Business Intelligence solution using Microsoft Excel. Through ETL processes, dimensional modeling, and analytical reporting, the solution transforms raw sales data into meaningful business insights. The resulting reports enable stakeholders to evaluate performance, identify growth opportunities, and make informed strategic decisions based on data.